

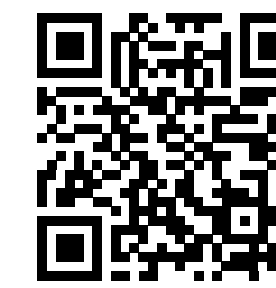


Test-Time Generation

A generative readout for fast-weight memory

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Paper



Website



Code

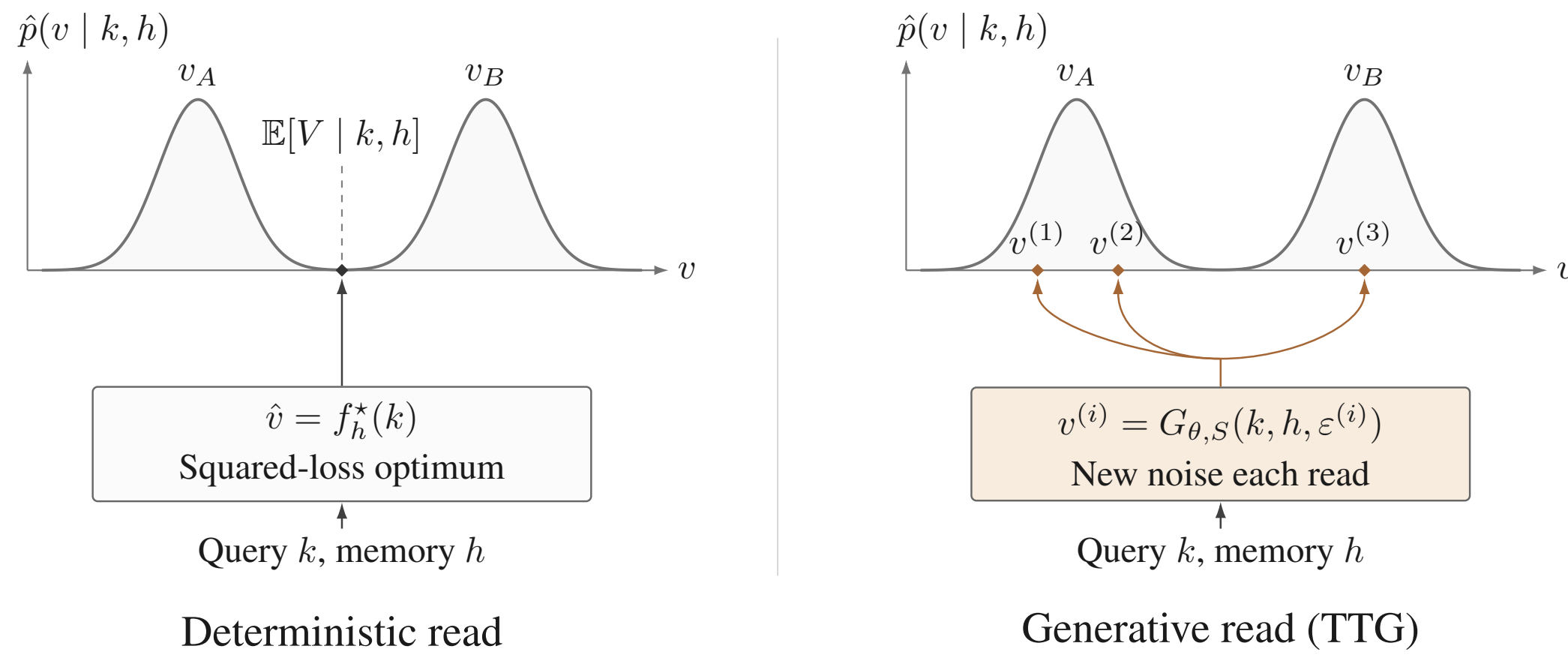
Motivation and approach

1 Deterministic reads can average answers

One query can match several values. Squared-error regression returns a conditional mean that can blend distinct answers.

2 Generative recall from the same memory

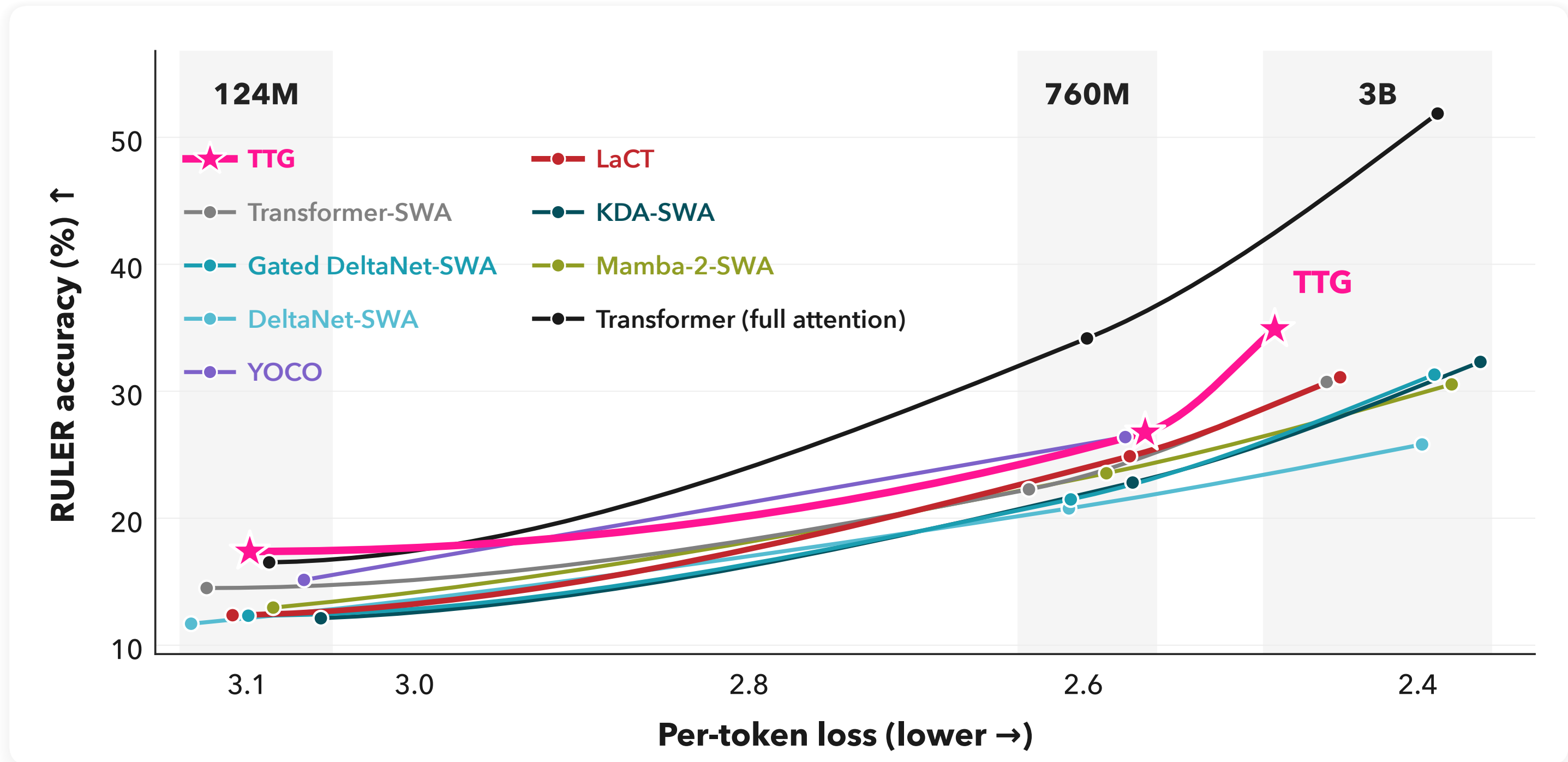
TTG generates from fresh noise, conditioned on the query, while keeping the fast-weight state fixed.



Idealized multi-step readout. A Bayes-optimal single step still returns the mean.

Long-context language modeling

124M / 760M / 3B families. TTG: $r = d = 1$, 4K-token chunks.
PTL: Book-3 30K–32K tail. RULER: 13 tasks, mean over 4K / 8K / 16K / 32K.



	124M		760M		3B	
Model	PTL ↓	RULER ↑	PTL ↓	RULER ↑	PTL ↓	RULER ↑
Transformert	3.0869	16.52	2.5978	34.16	2.3881	51.87
YOCO†	3.0661	15.14	2.5749	26.39	—	—
Transformer-SWA	3.1243	14.50	2.6325	22.27	2.4544	30.73
Gated DeltaNet-SWA	3.0994	12.32	2.6075	21.48	2.3900	31.31
DeltaNet-SWA	3.1336	11.69	2.6086	20.77	2.3974	25.81
KDA-SWA	3.0560	12.12	2.5705	22.81	2.3625	32.31
Mamba-2-SWA	3.0845	12.96	2.5862	23.54	2.3797	30.54
LaCT	3.1088	12.36	2.5722	24.88	2.4464	31.1
TTG	3.0985	17.38	2.5630	26.76	2.4855	34.9

RULER gain over LaCT: +5.02 (124M) / +1.88 (760M) / +3.8 (3B)

Bold: best bounded-state value. † Growing-cache reference. Conditioning changes total parameters (124M-family TTG: 172.7M). PTL and retrieval need not improve together.

Conditioning and parameter sharing

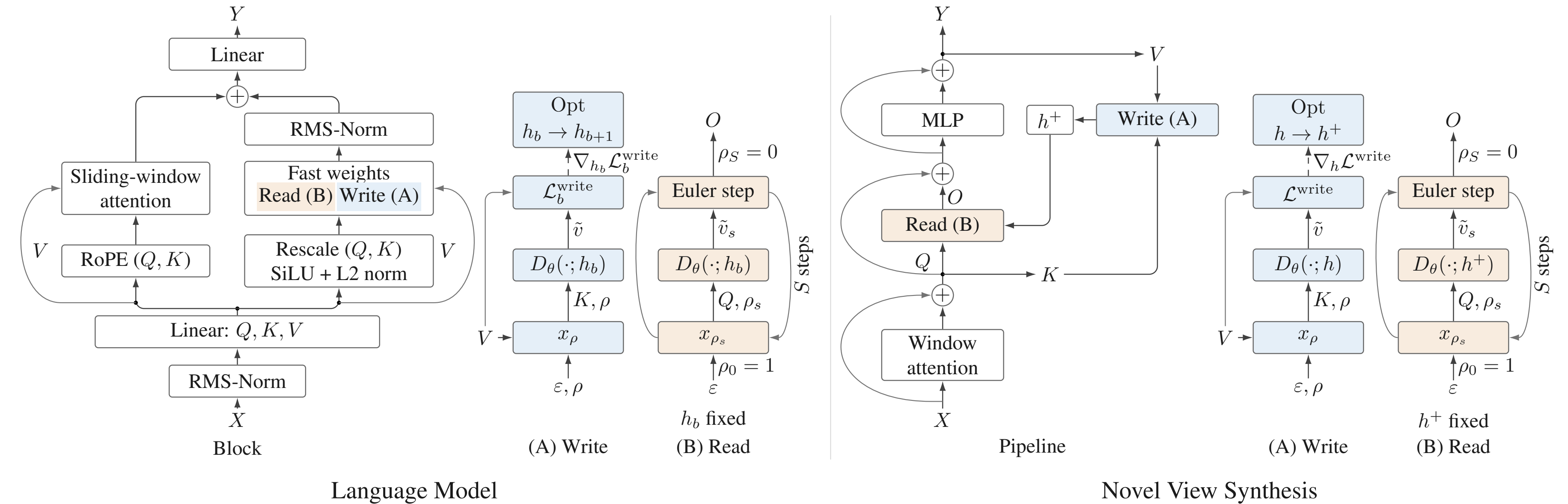
9.2M fewer NVS parameters (–15.5%) with shared AdaLN

Conditioning	Params (M)	GSO ↑	DL3DV ↑
Per-block AdaLN	59.3	24.351	16.392
Shared AdaLN	50.1	24.338	16.474
Shared + Reuse	50.1	24.538	16.431

TE on, continuous noise, $r = d = 1$; PSNR in dB.
This conditioning ablation uses a separate configuration from the main comparison.

Similar NVS quality.
32.5M fewer LM parameters.
Reuse shares modulation.

TTG: online writes, generative reads



Read prefix memory before the current chunk update.

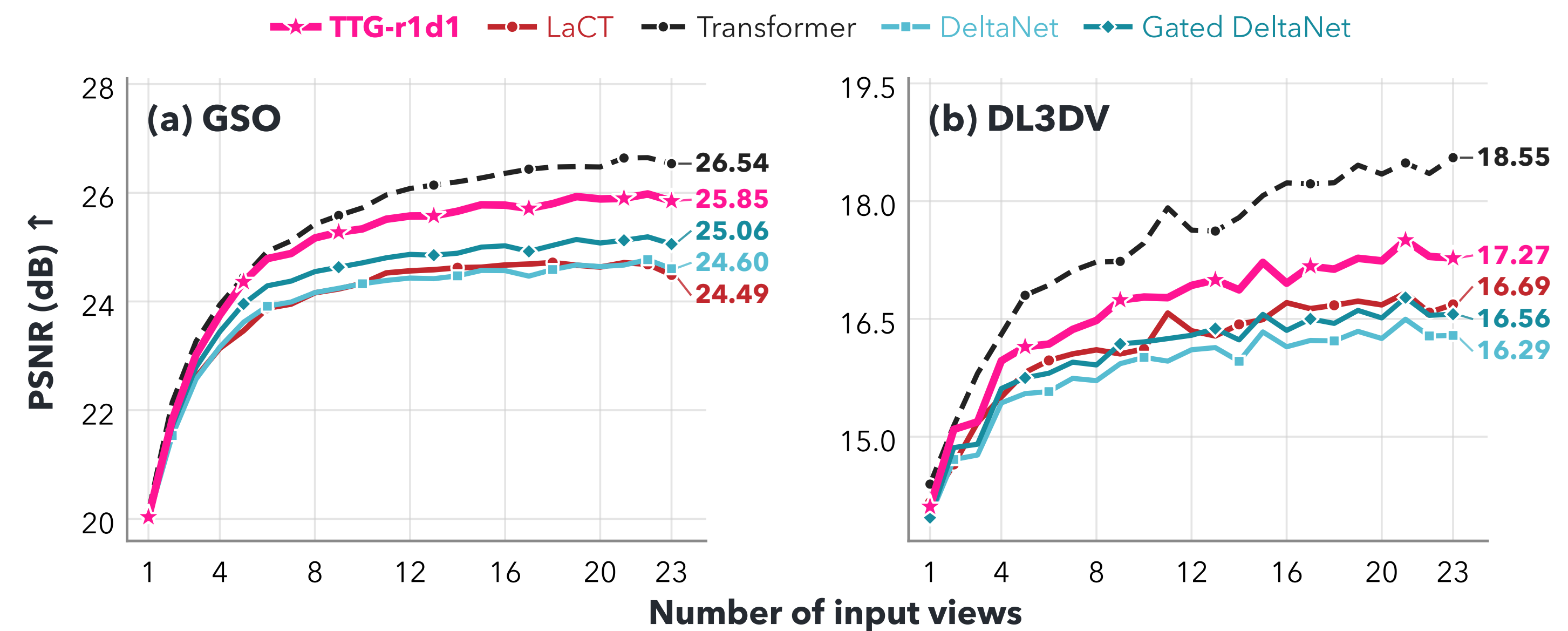
Write observations, then read the updated scene state.

TTG improves retrieval and view synthesis with fixed-size fast-weight memory.

Slow parameters stay fixed. r = write repeats; d = trained readout depth; S = evaluation depth.

Novel view synthesis

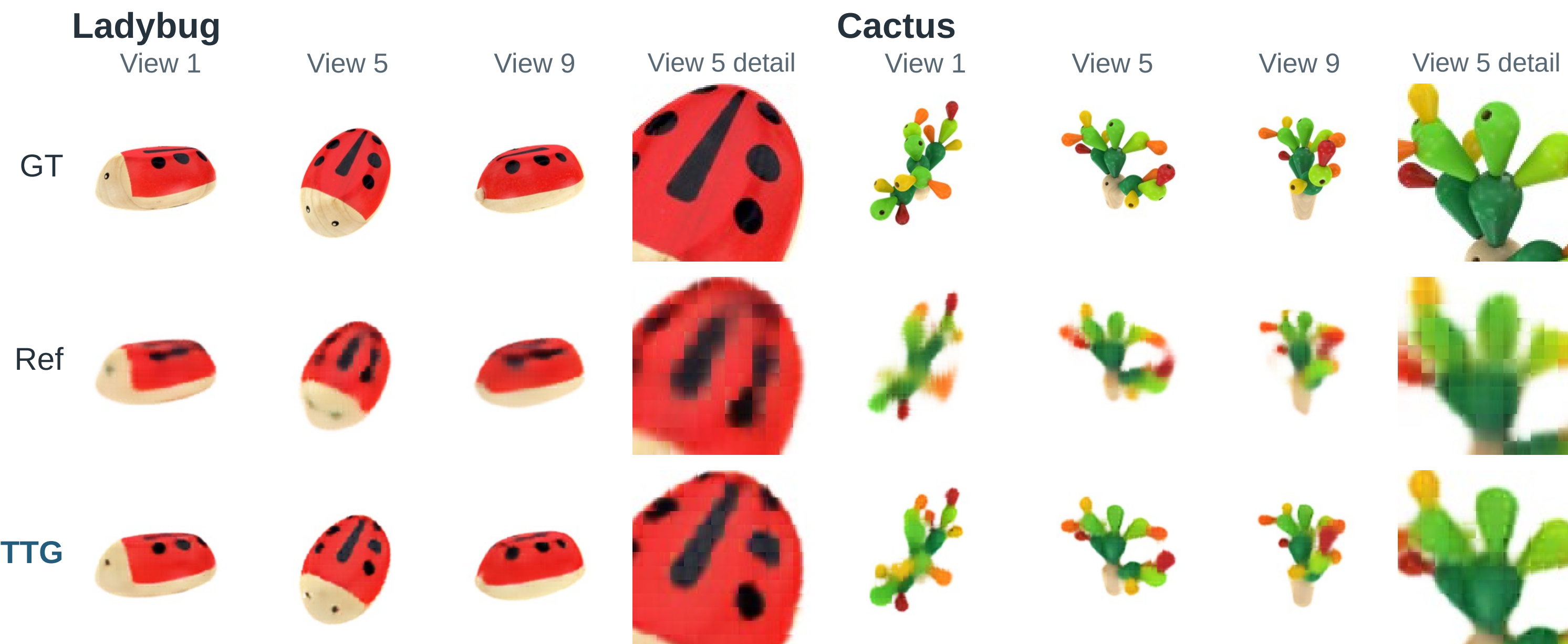
Objaverse → GSO and DL3DV. 12 training / 24 evaluation views.
1,019 GSO objects and 140 DL3DV scenes. One generated sample.



+0.94 dB on GSO / +0.45 dB on DL3DV over LaCT

Gain: mean over 23 positions. Curve labels: final target. Transformer has a growing cache.
LaCT/Transformer use a different view seed. PSNR alone does not test mean collapse.

Novel view synthesis: visual examples



Ladybug +3.75 dB / Cactus +3.43 dB over the deterministic reference

Views 1, 5, 9 + view-5 detail. All 24 views written first, including observed views.
Gains average all 24 targets for these objects. This differs from next-view prediction.